ADITHYA BHAT

Visa Research, Visa Inc., CA

◇ Email - dth.bht@gmail.com ◇ GitHub - https://github.com/adithyabhatkajake

◇ Website - https://adithyabhatkajake.github.io

RESEARCH INTERESTS

My interests are *Byzantine fault-tolerant distributed systems* and *applied cryptography*. My work aims at developing and evaluating cryptographic solutions for secure, fault-tolerant distributed systems. My current research focuses on energy-efficient consensus protocols in network settings such as synchronous, asynchronous, and all intermediate models; fault-tolerant cryptographic protocols such as PVSS and Random Beacons and Private set intersection.

EDUCATION

Purdue University, West Lafayette *(2018-2023)*
Advisor: Aniket Kate
Ph.D., Department of Computer Science

National Institute of Technology Karnataka, Surathkal *(2014-2018)*
Bachelor of Technology, Information Technology

WORK EXPERIENCE

Visa Research, Foster City (Aug 2023 - Present)
Staff Research Scientist *(Anderson Nascimento)*

Visa Research, Palo Alto (May 2022 - Aug 2022)
Ph.D. Research Intern *(Mahdi Zamani)*

- Developed *FastSync*: an efficient blockchain synchronization protocol
- Developed an efficient partially synchronous sharding protocol and implemented Instachain

VMware Research, Remote (May 2021 - Aug 2021)
Research Intern *(Alin Tomescu, Ittai Abraham)*

- Built a prototype of anonymous token system using Concord-BFT
- Developed *quick-pay*: a one-round trip low-latency payment system
- Developed a two-phase lock-free sharding solution using quick-pay

Purdue University, West Lafayette (Aug 2018 - Aug 2023)
Graduate Research Assistant *(Aniket Kate)*

- Researching energy efficient Byzantine fault tolerant consensus protocols.
- Developed mathematical models for protocol optimization to improve energy efficiency.
- Implemented, evaluated, and simulated cryptographic and distributed system protocols.

Indian Statistical Institute, Kolkata (July 2017 - December 2017)
Undergraduate Research Fellow *(Sushmitha Ruj)*

- Designed a storage auditing library based on compact proofs of retrievability.
- Implemented new transactions on an Ethereum client to build a publicly verifiable data-storage system.

Morgan Stanley, Bangalore (May 2017 - July 2017)

Software Analyst

- Worked on evaluating an elastic-search visualization plugin for the LevelDB database.

Indian Institute of Science, Bangalore (May 2016 - July 2016)

Indian Academy of Sciences Summer Research Fellow *(C. E. Veni Madhavan)*

- Evaluated Pollard's rho, William's $p+1$, $p-1$ factorization, and elliptic curve factorization methods for efficient batch factorization of numbers generated during the sieving phase of GGNFS.

PUBLICATIONS

1. *Lite-PoT: Practical Powers-of-Tau Setup Ceremony.* **CCS 2025.** Lucien K. L. Ng, Pedro Moreno-Sanchez, Mohsen Minaei, Panagiotis Chatzigiannis, **Adithya Bhat**, Duc V. Le. eprint
2. *Synchronous Distributed Key Generation without Broadcasts.* **Journal of Cryptology 2024**. Nibesh Shrestha, **Adithya Bhat**, Kartik Nayak, Aniket Kate. journal eprint
3. *SensorBFT: Fault-Tolerant Target Localization using Voronoi Diagrams and Approximate Agreement.* **ICDCS 2024**. Akhil Bandarupalli, **Adithya Bhat**, Somali Chaterji, Michael K. Reiter, Aniket Kate, Saurabh Bagchi. conference eprint
4. *Delphi: Efficient Asynchronous Approximate Agreement for Distributed Oracles.* **DSN 2024**. Akhil Bandarupalli, **Adithya Bhat**, Saurabh Bagchi, Aniket Kate, Chen-Da Liu-Zhang, and Michael K. Reiter. conference eprint
5. *HashRand: Efficient Asynchronous Random Beacon without Threshold Cryptographic Setup.* **CCS 2024**. Akhil Bandarupalli, **Adithya Bhat**, Saurabh Bagchi, Aniket Kate, and Michael Reiter. conference eprint
6. *Attacking and Improving the Tor Directory Protocol.* **IEEE S&P 2024**. Zhongtang Luo, **Adithya Bhat**, Kartik Nayak, and Aniket Kate. conference eprint
7. *EESMR - Energy Efficient State Machine Replication.* **Middleware 2023**. **Adithya Bhat**, Akhil Bandarupalli, Manish Nagaraj, Saurabh Bagchi, Aniket Kate, Michael Reiter. conference eprint
8. *The unique chain rule and its applications.* **Financial Cryptography 2023**. **Adithya Bhat**, Akhil Bandarupalli, Saurabh Bagchi, Aniket Kate, Michael Reiter. pre-conference conference eprint code
9. *OptRand - Optimistically Responsive Reconfigurable Distributed Randomness.* **NDSS 2023**. **Adithya Bhat**, Nibesh Shrestha, Aniket Kate, Kartik Nayak. conference eprint protocol code crypto code video
10. *OpenSquare: Decentralized Repeated Modular Squaring Service.* **CCS 2021**. Sri Aravinda Krishnan Thyagarajan, Tiantian Gong, **Adithya Bhat**, Aniket Kate, Dominique Schroder. conference eprint code
11. *RandPiper – Reconfiguration Friendly Random Beacons with Quadratic Communication.* **CCS 2021**. **Adithya Bhat**, Nibesh Shrestha, Aniket Kate, Kartik Nayak. conference eprint code
12. *Reparo - Publicly Verifiable Repair Layer for any Blockchain.* **FC 2021**. Sri Aravinda Krishnan Thyagarajan, Adithya Bhat, Bernardo Magri, Daniel Tschudi, Aniket Kate. conference eprint
13. *Verifiable Timed Signatures for Blockchains.* **CCS 2020**. Sri Aravinda Krishnan Thyagarajan, Adithya Bhat, Guilio Malavolta, Nico Dottling, Aniket Kate, Dominique Schroder. conference eprint code

PATENTS

1. *Accountable Decentralized Anonymous Payments.* United States Patent Application 20240265373. patent.
2. *Fast sync blockchain system and method.* United States Patent Application WO2024026321A1. patent.
3. *A system for partially synchronous scalable blockchains.* United States Patent Application WO2024233453A1. patent

TECH REPORTS

1. Ira: Efficient Transaction Replay for Distributed Systems. **Adithya Bhat**, Harshal B Shah, Mohsen Minaei. eprint
2. FPS: Flexible Payment Systems. **Adithya Bhat**, Srinivasan Raghuraman, Panagiotis Chatzigiannis, Duc V Le, Mohsen Minaei. eprint
3. SoK: *Fully-homomorphic encryption in smart contracts.* **Poster - FHE.org workshop**. Daniel Aronoff, **Adithya Bhat**, Panagiotis Chatzigiannis, Mohsen Minaei, Srinivasan Raghuraman, Robert M. Townsend, Nicolas Xuan-Yi Zhang. eprint
4. *Leto - Partially Synchronous Unique Chains made flexible.* **Adithya Bhat**, Saurabh Bagchi, Aniket Kate, Michael Reiter. code
5. *UTT: Decentralized Ecash with Accountable Privacy.* **Science of Blockchain Conference 2023**. Alin Tomescu, Adithya Bhat, Benny Applebaum, Ittai Abraham, Guy Gueta, Benny Pinkas, and Avishay Yanai. eprint code
6. *Transitive Network - A Tokenless IOU-based Credit Network.* **Cryptocurrency Implementers Workshop, FC 2019. Adithya Bhat**, Pedro Moreno Sanchez, Aniket Kate. code

SOFTWARE ARTIFACTS

1. Developed a synchronous networking library to implement SMR protocols. code (Rust)
2. Implemented Apollo [8] (protocol node, normal client and special client) using the Rust networking library. code (Rust)
3. Implemented Sync HotStuff (normal protocol node, round robin protocol node, client) using the Rust networking library. code (Rust)
4. Developed a plug-and-play framework using libp2p to run and simulate distributed system protocols. The framework provides interfaces to aid faster prototyping of distributed system protocols. code (Go-lang)
5. Implemented Sync HotStuff using the go networking library. code (Go-lang)
6. Implemented Apollo [8] using the go networking library. code (Go-lang)
7. Implementation of E2C [7]. code (C++)
8. Developed a linearly homomorphic time-lock puzzle library. code (C)

TALKS

1. Attacking and Improving the Tor Directory Protocol. *(RWC 2025)*

2. A study of practical BFT systems. *(Supra Team)*
3. A study of practical BFT systems. *(Espresso Systems)*
4. A study of practical BFT systems. abstract *(Next Generation Secure Distributed Computing Seminar, Dagstuhl)*
5. OptRand, an optimistically responsive distributed random beacon protocol. video *(Web3 Randomness Workshop)*
6. Reconfiguration-friendly Byzantine Fault-tolerant Distributed randomness. slides *(KU Leuven)*
7. Unique Chain Rule and its applications. slides *(FC 2023)*
8. Reconfiguration-friendly Byzantine Fault-tolerant Distributed randomness. slides *(Boston University)*
9. Flexible State Machine Replication. slides *(Midwest Crypto Day - Lightning session)*
10. OptRand - Optimistically Responsive Reconfigurable Distributed Randomness. video *(NDSS 2023)*
11. FastSync: Using the future to verify the past. video *(CESC 2022)*
12. RandPiper - Reconfiguration friendly random beacons with quadratic communication. *(CCS 2021)*
13. Reparo - Publicly Verifiable Repair Layer for any blockchain. video *(FC 2021)*
14. Transitive network - A tokenless IOU-based Credit Network. Cryptocurrency Implementers Workshop. *(FC 2019)*

ACADEMIC SERVICE

- Program Committee:
  - 2026: FC, CCS, ACNS, Usenix Security
  - 2025: FC, CCS, CVC
  - 2024: CCS
- Reviewer for
  - 2024: ACM TOPS, Journal of Cryptology
  - 2023: SOSP AEC, IET
- External Reviewer for
  - 2025: ITCS, Crypto
  - 2024: IEEE S & P
  - 2023: CCS, IET, Middleware, IEEE S & P
  - 2022: CCS, PODC, IEEE S & P, CESC
  - 2021: AFT, FC, PODC, IEEE S & P
  - 2020: Usenix Security, IEEE S & P
  - 2019: NDSS